



Fig. 7: Antennae of AN/SPY-1 and AN/SPG-62 radars in Aegis cruiser (left) and Aegis destroyer (right)

For complex missions, entire VLS could be loaded with SM-2 missiles.

AN/SPG-62 illumination/fire-control radars. This radar operates in I/J NATO frequency band (8GHz-20GHz) with a peak power of approximately 10kW. It is a continuous-wave illumination radar and does not chirp out signals as pulses. Instead, it focuses its energy as a fine pencil beam on the target, like focusing a flashlight on an object. Due to this, this radar is called an illumination radar.

In military parlance, this is known as painting the target. This radar is mechanically turned to steer the beam. There are four such radars in Aegis cruiser; two are rearward-facing and two are forward-facing. This arrangement provides 360° coverage. With each radar illuminating a target, these radars can illuminate up to four targets.

Aegis destroyers have three of these radars and the absence of AN/SPN-49 radar leaves it with lower air defence capability. This makes Aegis cruisers a prime node in the air defence architecture of CSG defending against cruise missile strikes.

Typical engagement sequence

In the engagement sequence, first, incoming target is detected through AN/SPN-49 surveillance radar. Aegis C² system directs that target to be

tracked by AN/SPY-1 radar. From this tracking, the target's trajectory is derived in Aegis C² system and the target's future course is computed. Then, the point of interception is calculated, which is a place where the target can reach in a few moments. SM-2 SAM is launched to reach this interception point through midcourse guidance.

Due to the very long range involved, the parent vessel gives commands for the missile to fly, which is called midcourse guidance. These commands are supplied through a datalink, typically a radio channel. The missile receives this information through the datalink and flies accordingly. AN/SPY-1 radar provides this datalink too.

By the time SM-2 approaches incoming target, Aegis C² system directs the fire-control radar to illuminate the incoming target. When the missile reaches an optimum distance from the target, it starts using its radar receiver and guides itself towards the target.

AN/SPY-1 allows many such beams to be formed. Because of this, many datalink channels can be established so that many SM-2 missiles can be given guidance commands. This enables many missiles to stay on air to intercept the same target or multiple targets.

Similarly, Aegis C² system time

shares illumination radars for a large number of targets to neutralise simultaneous cruise-missile strikes. The number of targets that could be engaged through this feature is kept as classified information. But without this time-sharing feature, the number of targets intercepted will be equal to the number of fire-control radars.

Cooperative engagement capability, the latest trend

As the CSG's cruise missile defences increased in capability, the Soviets fielded faster cruise missiles; some of these cruise missiles were even capable of covering a kilometre in a second. Because of this high speed, reaction time for the CSG decreased to a great extent. Windows of interceptions also reduced for the CSG against these faster cruise missiles.

The Soviets also increased the range of cruise missiles. They augmented the number of cruise missiles carried by cruise-missile-carrying platforms. Soviet submarines, surface vessels and bombers were able to carry more cruise missiles in a compact manner. So, more number of cruise missiles of lethal speed could be fired from longer range and also in greater numbers.

Though Soviet Union is defunct now, Russian Federation and China have this cruise missile strike capability. To defend from this renewed attack of fury, American CSGs needed a new deal in cruise missile defence.

Cooperative engagement capability (CEC) is the new sinister air defence scheme of USN. Ironically, CEC has not been achieved through addition of any new radars or weapon systems but by sharing these. Through this sharing, the capability level of air defence has increased exponentially. This is the first step towards a futuristic warfare called network centric warfare (NCW).

In CEC, tracking data from radars present in the destroyers and cruisers is shared using identical algorithms. Each ship tracks a target with a radar and, simultaneously, sends this data